$1 + 12 - 14$

$1 : -4$

$11 + 12 + 10|$

$4 + 1 \le 3 + 3$

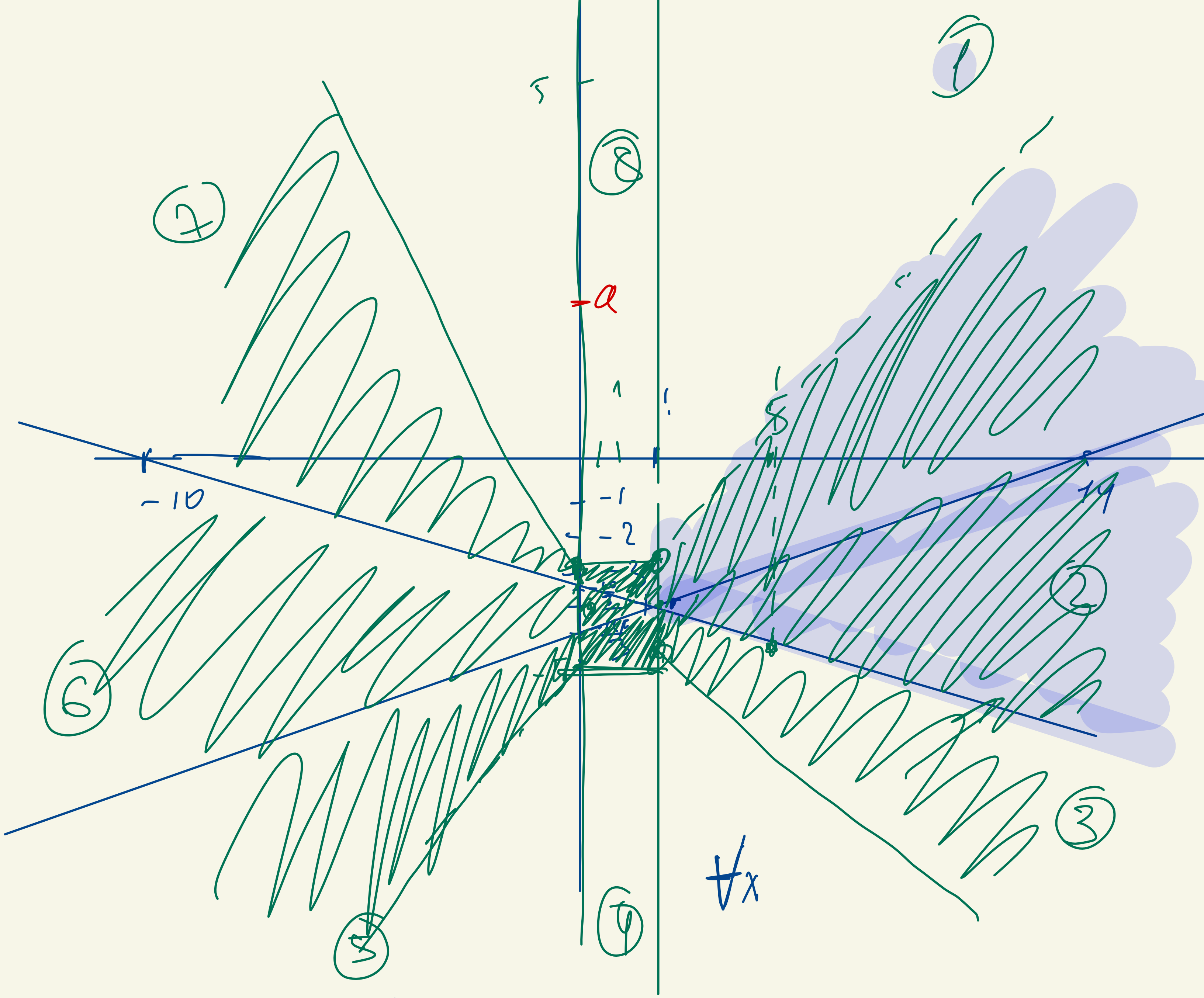
$-x - \cancel{3a} - 10 - x + \cancel{3a} + 14$

$\le \cancel{3x} - \cancel{3x} + 6$

$-2x + 4 \le 6$

$-2x \le 2$

$x \ge -1$



$|x + 3a + 10| + |x - 3a - 14| \le 3|x| + 3|x - 2|$

1) $x + \cancel{3a} + 10 + x - \cancel{3a} - 14 \le 3x + 3x - 6$

$\cancel{x} + 3a + 10 - \cancel{x} + 3a + 14 \leq 3x + 3x - 6$

$6a + 24 \leq 6x - 6$

$a \leq x - 5$

(2) $a \geq \frac{-10 - x}{3}$ $\qquad x - 3a - 14 \geq 0$

$a \leq \frac{x - 14}{3}$

$x + 3\cancel{a} + 10 + x - 3\cancel{a} - 14 \leq 3x + 3x - 6$

$2x - 4 \leq 6x - 6$

$4x \geq 2$

$x \geq \frac{1}{2}$

$x + 3a + 10 \leq 0$ $\qquad x - 3a - 14 \geq 0$

$a \leq \frac{-10 - x}{3}$ $\qquad a \leq \frac{x - 14}{3}$

$-\cancel{x} - 3a - 10 + \cancel{x} - 3a - 14 \leq 3x + 3x - 6$

$-6a - 24 \leq 6x - 6$

$-6a \leq 6x + 18$

$a \geq -x - 3$ ✓

(4) $-6a-24 \le 3x - 3x + 6$

$-6a \le 30$

$a \geq -5$

$-6a - 24 \le -3x - 3x + 6$

$-6a \le -6x + 30$

$a \geq \frac{6x - 30}{6} = x - 5$

$- x - \cancel{3a} - 10 \quad - x + \cancel{3a} + 14 \quad \le -3x - 3x + 6$

$-2x + 4 \le -6x + 6$

$4x \le 2$

$x \le \frac{1}{2}$

(7) $x + 3a + 10 \quad - x + 3a + 14 \le -3x - 3x + 6$

$6a + 24 \le -6x + 6$

$a \le \frac{-6x - 18}{6} = -x - 3$

$6a + 24 \le 3x - 3x + 6$

$6a \le -18$

$a \le -3$

9. $-x - 3a - 10 \quad - x + 3a + 14 \le 3x \quad 3x + 6$

$$-2x + 4 \geq 6$$

$$2x \geq -2$$

$$x \geq -1$$

$$x + 3a + 10 < 0$$

$$a < \frac{-10 - x}{3}$$

$$x > 0$$

$$x - 3a - 14 < 0$$

$$a > \frac{x - 14}{3}$$

$$x - 2 < 0$$

$$-3x + 6$$

$$-x - 3a - 10 - x + 3a + 14 \leq 3x - 3x + 6$$

$$-2x + 4 \leq 6$$

$$-2x \leq 2$$

$$x \geq -1$$

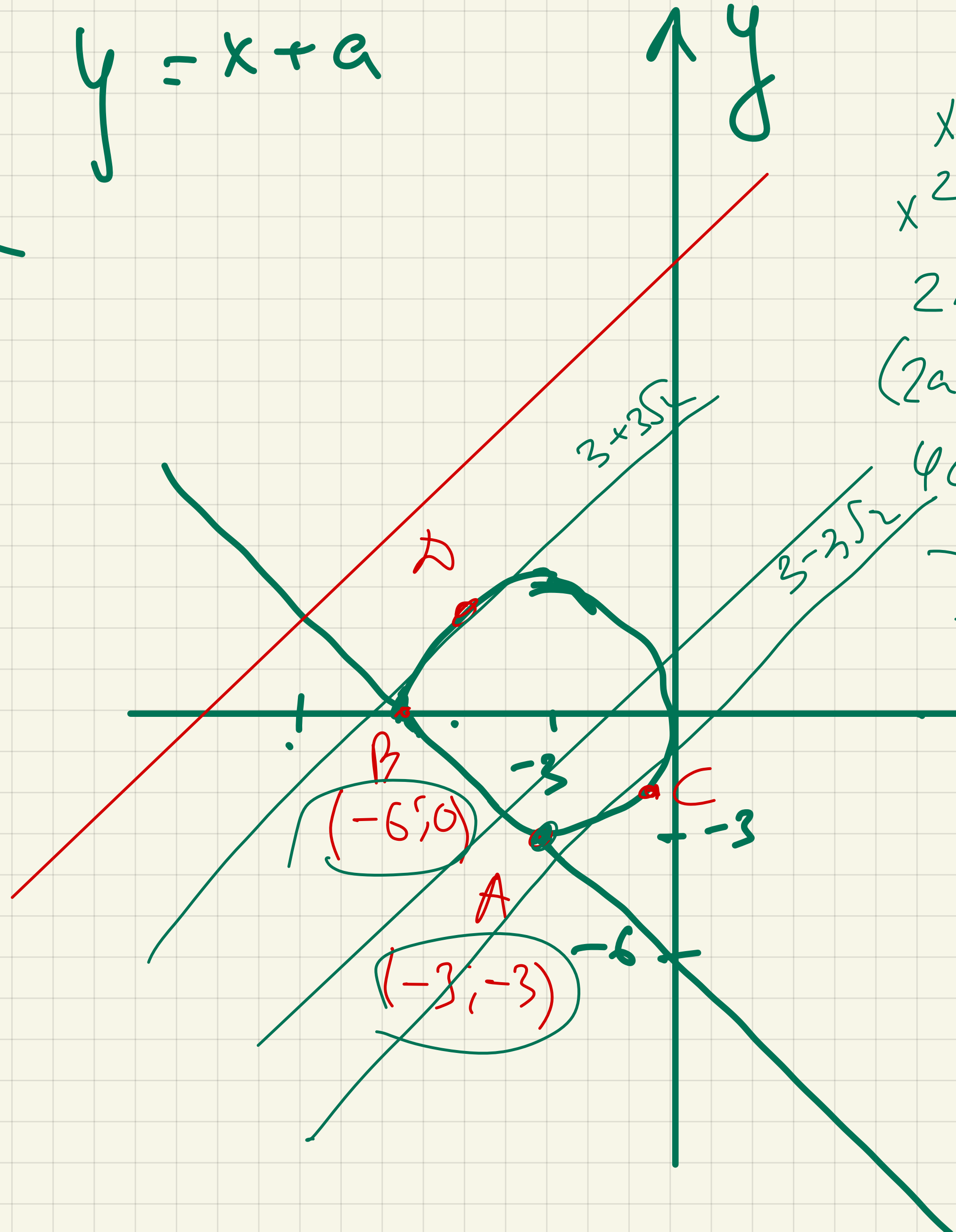
$$\begin{cases} (x^2+y^2+6x)\sqrt{x+y+6} = 0 \\ y = x+a \end{cases}$$

2 реш.
ра

$$\begin{cases} \left[\begin{cases} x^2+y^2+6x=0 \\ x+y+6 \geq 0 \end{cases}\right. \\ y = x+a \end{cases}$$

$(x+3)^2+y^2=9$

$y \geq -x-6$



$x^2+(x+a)^2+6x=0$

$x^2+x^2+2ax+a^2+6x=0$

$2x^2+x(2a+6)+a^2=0$

$(2a+6)^2-4\cdot 2\cdot a^2=0$

$4a^2+48a+36-8a^2=0$

$-4a^2+24a+36=0$

$-a^2+6a+9=0$

$$\frac{-6\pm\sqrt{36+9\cdot 4}}{-2}$$

$$\frac{-6\pm 6\sqrt{2}}{-2}$$

$$\frac{-3\pm 3\sqrt{2}}{-1}$$

$a \in \{3-3\sqrt{2}\} \cup [0;6] \cup \{3+3\sqrt{2}\}$ $3 \pm 3\sqrt{2}$

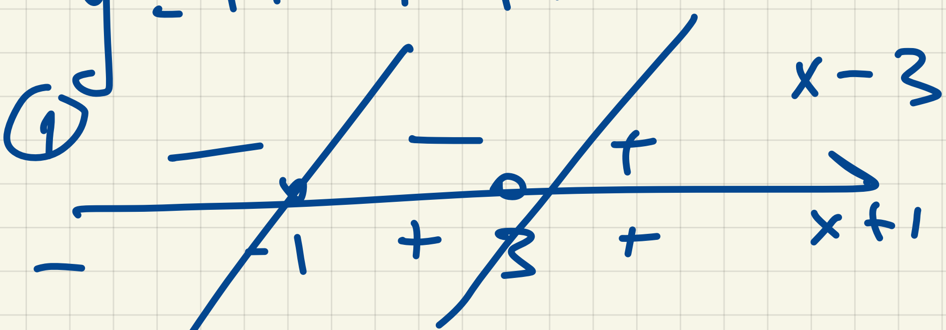
$$\begin{cases} (|x+1| + |x-3| - y)\sqrt{10-x-y} = 0 \\ y = x + a \end{cases}$$

$10 - x - y \geq 0$

$y \leq 10 - x$

2 реш ?

$y = |x+1| + |x-3|$



$x \leq -1$

$-x-1-x+3 = y$

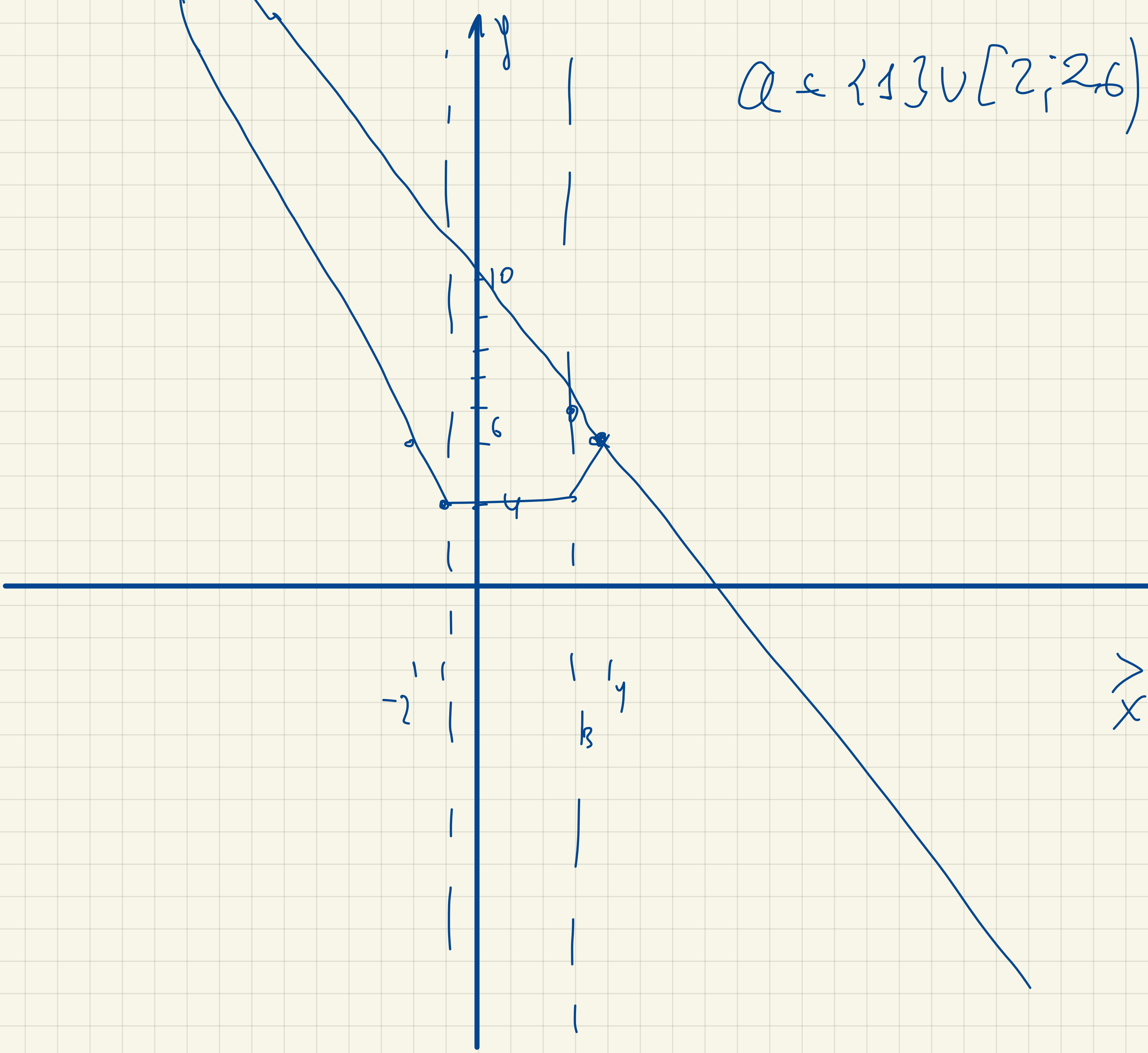
$-2x+2 = y$

$x+1-x+3 = y$

$x+1+x-3 = y$

$2x-2 = y$

$a \in \{1\} \cup [2; 2{,}6)$



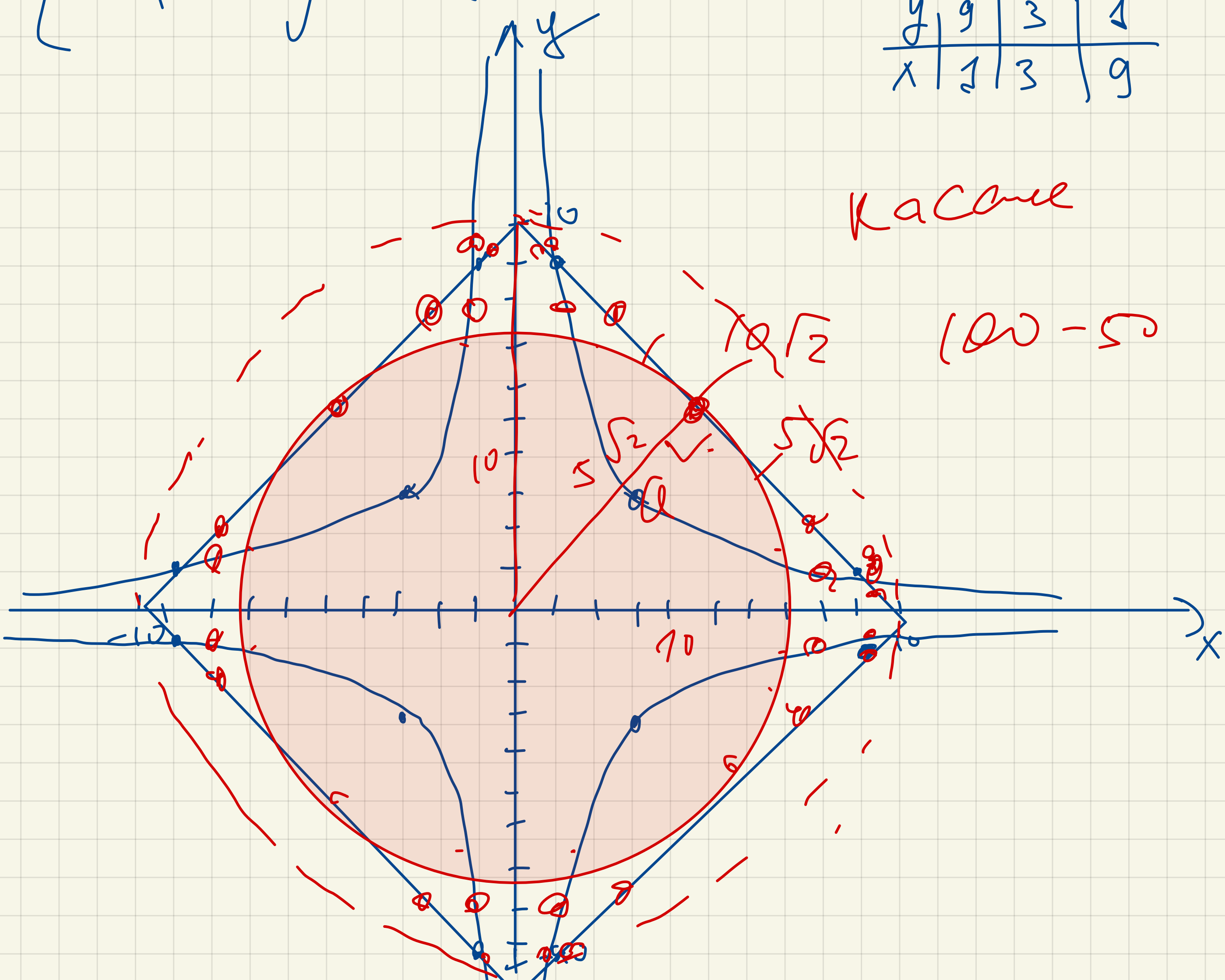
$$\begin{cases} (|x|+|y|-10)(9-|xy|)=0 \\ x^2+y^2=a^2 \end{cases}$$

12 реш?

$$\begin{cases} \left[\begin{array}{l} |x|+|y|-10=0 \\ |xy|=9 \end{array}\right. \\ x^2+y^2=a^2 \end{cases}$$

$$\left[\begin{array}{l} y=-\frac{9}{x} \\ y=\frac{9}{x} \end{array}\right.$$

y	9	3	1
x	1	3	9

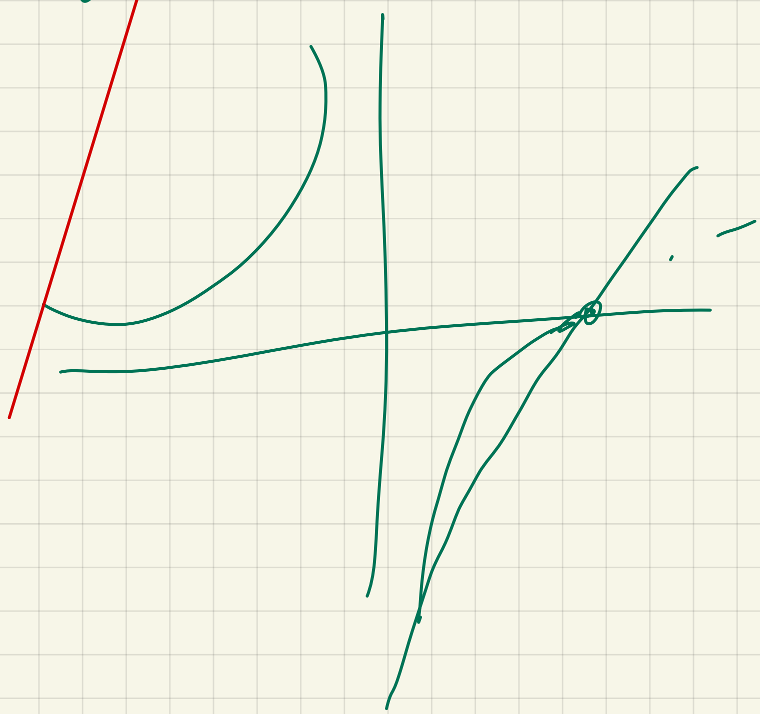


$a = 5\sqrt{2}$ или $a = 10$

$$\begin{cases} (xy - 2x + 12)\sqrt{y - 2x + 12} = 0 \\ y = 3x + a \end{cases}$$

2 разных решения

$y \geq 2x - 12$



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